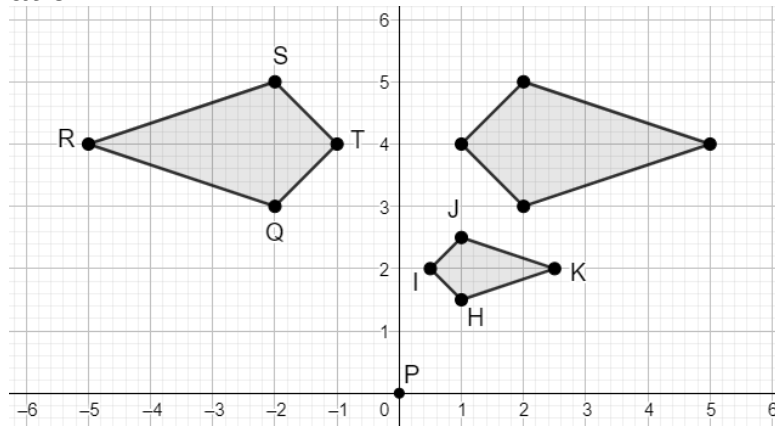
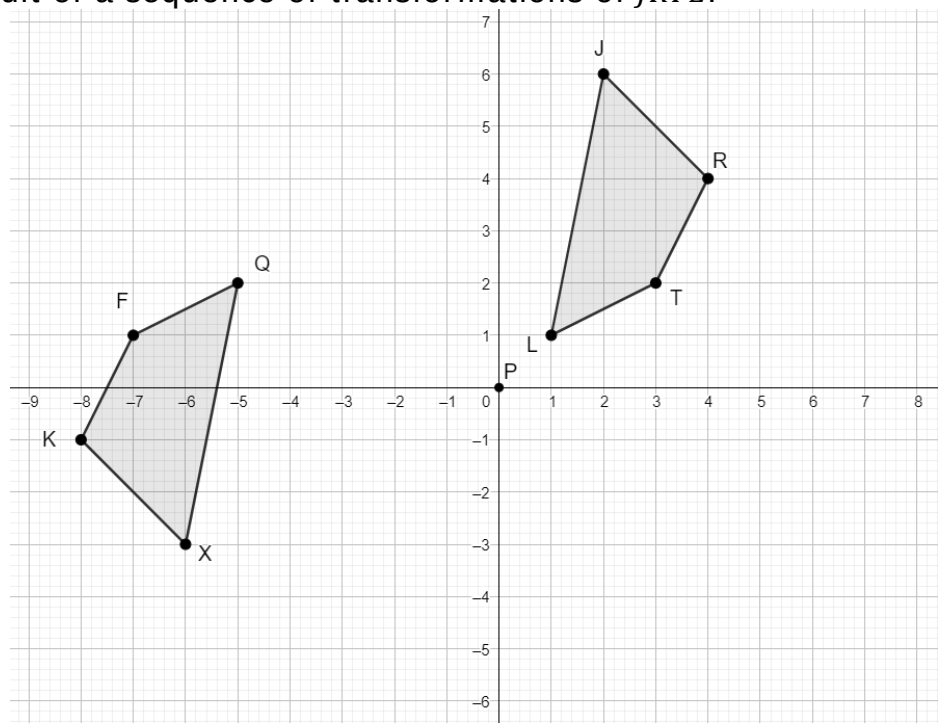


$HIJK$ is the result of a sequence of transformations of $QTSR$. The unlabelled quadrilateral is the first transformation.



- Describe the first transformation of $QTSR$.
Reflection over y -axis
- Write the transformation algebraically.
 $(x, y) \rightarrow (-x, y)$
- Describe the second transformation that results in $HIJK$.
Dilation with a scale factor of $\frac{1}{2}$ centered at $(0, 0)$
- Write the second transformation algebraically.
 $(x, y) \rightarrow \left(\frac{1}{2}x, \frac{1}{2}y\right)$
- Explain if the sequence of transformations preserves distance.
The first transformation does preserve distance, but the second transformation does not.

$XKFQ$ is the result of a sequence of transformations of $JRTL$.



Julia and Nathan were determining the sequence of transformations, but each came up with a different sequence. Complete the table for the missing portions of their transformations.

			First Transformation	Second Transformation
6.	Julia's Sequence	Written Description	Translate right 4 and down 3	Rotate 180° about the origin.
7.		Algebraic Description	$(x,y) \rightarrow (x + 4,y - 3)$	$(x,y) \rightarrow (-x,-y)$
8.	Nathan's Sequence	Written Description	Rotate 180° about the origin.	Translate right 4 and down 3
9.		Algebraic Description	$(x,y) \rightarrow (-x,-y)$	$(x,y) \rightarrow (x + 4,y - 3)$

10. Considering both proposed sequences, you conclude that correct.

- ✖ Julia's sequence is
- ☐ Nathan's sequence is
- ☐ Both sequences are